

Axisymmetric Navier-Stokes with swirl: the $1/r^4$ axis term (without Φ -renormalization)

Status: OPEN obstruction. Not a Clay proof. Not ChatVault. This is the operator before the algebraic cancel.

Hunt note. Zenodo public API (2026-08-26): 22050974, 22050975, 22045467, 21071991, 20405405, and 20405597 all introduce Φ -renormalization. No separate public “without cancel” PDF was found. Domain Architect therefore states the pre-cancel operator here and points at the later live face.

Operator (still in the equations)

Axisymmetric-with-swirl, cylindrical (r, θ, z) . Angular momentum $\Gamma = r u_\theta$. Renormalized azimuthal vorticity $\Omega = \omega_\theta / r$. The swirl-vorticity source still carries the centrifugal axis term:

$$D_t \Omega = (1/r^4) \partial_z (\Gamma^2) + \nu L_{\text{cyl}} \Omega$$

with $D_t = \partial_t + u^r \partial_r + u^z \partial_z$ and L_{cyl} the weighted cylindrical Laplacian on Ω . Equivalent source form: $(1/r) \partial_z (u_\theta)^2$ after dividing the azimuthal vorticity equation by r . As $r \rightarrow 0$ the $1/r^4$ prefactor is the classical axis obstruction. Standard H^1 estimates meet a Hardy inequality with weight r^{-4} .

What this face does not do

- It does not apply $\Phi = \Gamma/r^2 = u_\theta/r$.
- It does not use $r^{-4} \partial_z (\Gamma^2) = \partial_z (\Phi^2)$.
- It does not claim global regularity, Clay Statement B, or RH.

Later face (WITH cancel)

Live Φ -renormalization (Q_1 -augmented / Φ system, algebraic cancel of the $1/r^4$ term): DOI 10.5281/zenodo.22050974, file 01_phi_renormalization.pdf. Sibling 10.5281/zenodo.22050975. June conditional Φ : 10.5281/zenodo.21071991 (archive). That later face still does not claim unconditional classical 3D Navier-Stokes.

Domain Architect is a Functional Role Analysis classifier. Canonical SFE unresolved. This PDF is a desktop face so Jonathan can inquire the pre-cancel operator next to the live cancel paper.